Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

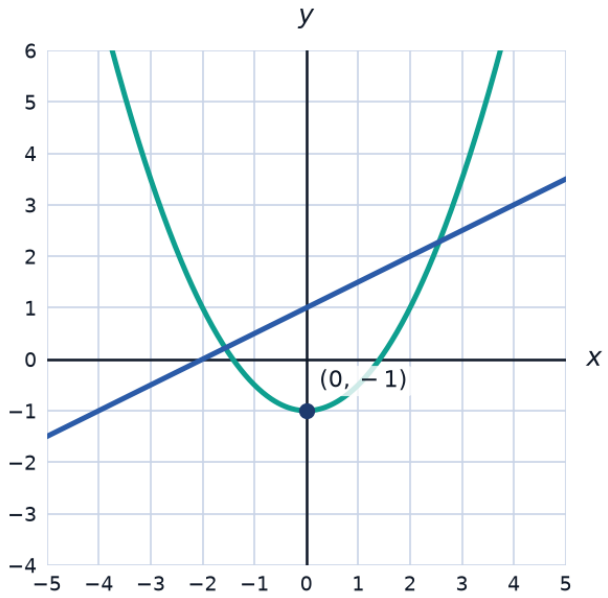
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Basic probability

Two fair six-sided dice are rolled. Find the probability their sum is 8.

2 [Foundational] • Target: Complement rule

If $P(E) = 0.35$, find $P(E')$.



3 [Foundational] • Target: Permutations and combinations

How many ways can a committee of 4 be chosen from 10 people?

4 [Foundational] • Target: Permutations and combinations

How many ways can 2 officers (president, vice president — order matters) be chosen from 6 candidates?

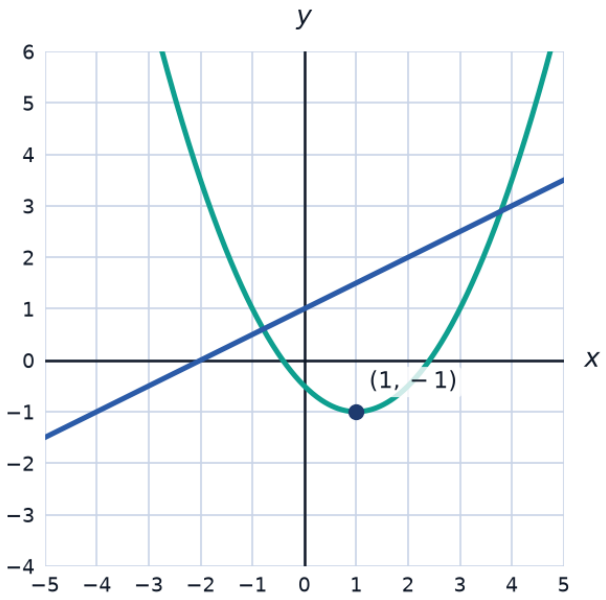
5 [Foundational] • Target: Permutations and combinations

Explain, using Theorem 6.1, how $\binom{6}{2}$ relates to $P(6, 2)$ from the previous question.

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6 [Foundational] • Target: Independent events and the addition rule

Events A and B are independent, with $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.



7 [Foundational] • Target: Independent events and the addition rule

A card is drawn from a standard 52-card deck. Find $P(\text{Queen or Spade})$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Combinatorial Probability & Geometric Probability

How many ways can a committee of 5 be chosen from 17 people? Show your complete algebraic work.

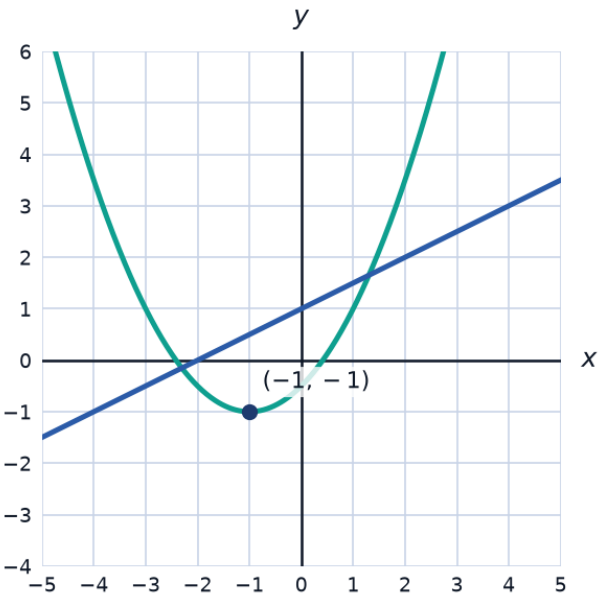
9 [Intermediate] • Target: Basic probability

How many ways can a committee of 2 be chosen from 18 people? Show your complete algebraic work.

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10 [Intermediate] • Target: Complement rule

How many ways can a committee of 6 be chosen from 18 people? Show your complete algebraic work.



11 [Intermediate] • Target: Permutations and combinations

How many ways can a committee of 2 be chosen from 14 people? Show your complete algebraic work.

12 [Intermediate] • Target: Independent events and the addition rule

How many ways can a committee of 5 be chosen from 11 people? Show your complete algebraic work.

13 [Intermediate] • Target: Discrete Random Variables & Variance

How many ways can 2 distinct positions be filled from 12 candidates, where order matters? Show your complete algebraic work.

14 [Intermediate] • Target: Continuous Distributions & Probabilities

How many ways can 4 distinct positions be filled from 9 candidates, where order matters? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

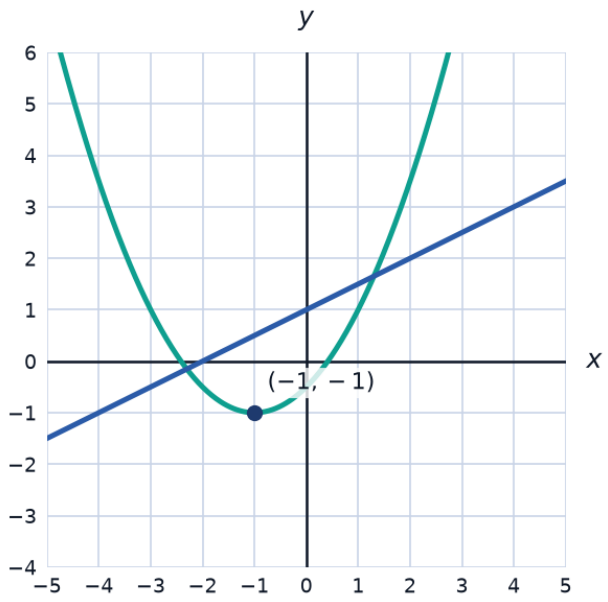
15 [Challenging] • Target: Conditional Probability & Joint Events • Cross-Lesson: 5.01 Introductory Calculus

How many ways can 4 distinct positions be filled from 14 candidates, where order matters? Show your complete algebraic work.

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16 [Challenging] • Target: Combinatorial Probability & Geometric Probability • Cross-Lesson: 1.01 Complex Numbers

How many ways can 2 distinct positions be filled from 11 candidates, where order matters? Show your complete algebraic work.



17 [Challenging] • Target: Basic probability • Cross-Lesson: 2.01 Sequences & Series

How many ways can 4 distinct positions be filled from 15 candidates, where order matters? Show your complete algebraic work.

18 [Challenging] • Target: Complement rule • Cross-Lesson: 3.01 Functions & Graphs

Events A and B are independent, with $P(A) = 0.1$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.

19 [Challenging] • Target: Permutations and combinations • Cross-Lesson: 4.01 Matrices & Vectors

Events A and B are independent, with $P(A) = 0.2$ and $P(B) = 0.3$. Find $P(A \text{ and } B)$.

20 [Challenging] • Target: Independent events and the addition rule • Cross-Lesson: 5.01 Introductory Calculus

Events A and B are independent, with $P(A) = 0.5$ and $P(B) = 0.1$. Find $P(A \text{ and } B)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Basic probability

A bag contains 9 yellow marbles, 5 red marbles, and 7 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.

a. $\frac{1}{3}$

b. $\frac{3}{7}$

c. $\frac{10}{21}$

d. $\frac{5}{21}$

22 [Foundational] • Target: Complement rule

A bag contains 9 yellow marbles, 9 red marbles, and 6 blue marbles. A marble is drawn at random. Find $P(\text{yellow})$.

a. $\frac{5}{12}$

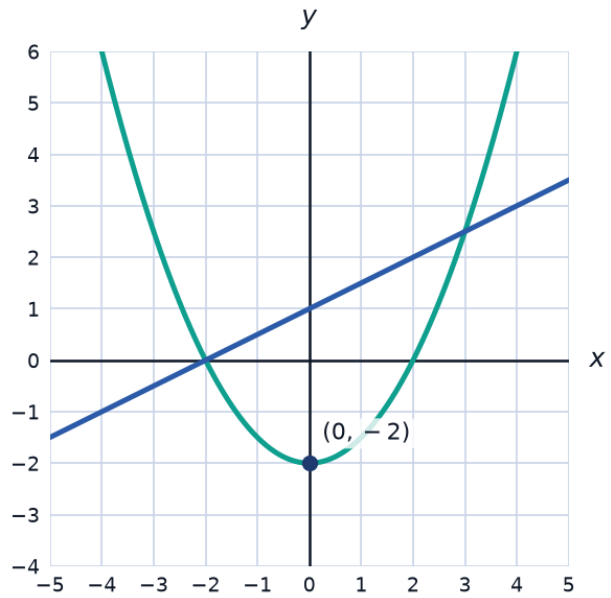
b. $\frac{11}{24}$

c. $\frac{1}{4}$

d. $\frac{3}{8}$

23 [Foundational] • Target: Permutations and combinations

A bag contains 7 blue marbles, 7 green marbles, and 3 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.



a. $\frac{9}{17}$

b. $\frac{8}{17}$

c. $\frac{3}{17}$

d. $\frac{7}{17}$

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24 [Foundational] • Target: Independent events and the addition rule

A bag contains 7 blue marbles, 7 green marbles, and 5 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.

a. $\frac{7}{19}$

b. $\frac{8}{19}$

c. $\frac{9}{19}$

d. $\frac{5}{19}$

25 [Foundational] • Target: Discrete Random Variables & Variance

A bag contains 4 blue marbles, 5 green marbles, and 5 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.

a. $\frac{2}{7}$

b. $\frac{1}{2}$

c. $\frac{3}{7}$

d. $\frac{5}{14}$

26 [Foundational] • Target: Continuous Distributions & Probabilities

A bag contains 9 blue marbles, 2 green marbles, and 4 yellow marbles. A marble is drawn at random. Find $P(\text{blue})$.

a. $\frac{4}{15}$

b. $\frac{2}{15}$

c. $\frac{2}{3}$

d. $\frac{3}{5}$

27 [Foundational] • Target: Conditional Probability & Joint Events

A bag contains 3 red marbles, 3 blue marbles, and 4 green marbles. A marble is drawn at random. Find $P(\text{red})$.

a. $\frac{3}{10}$

b. $\frac{1}{2}$

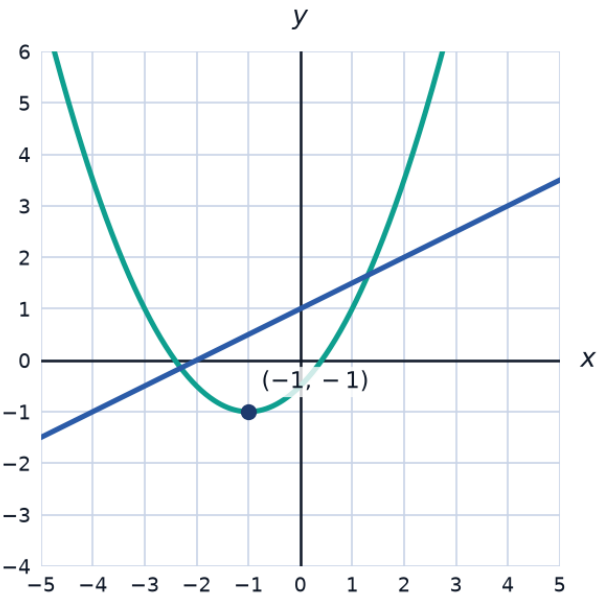
c. $\frac{3}{5}$

d. $\frac{2}{5}$

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28 [Foundational] • Target: Combinatorial Probability & Geometric Probability

How many ways can a committee of 6 be chosen from 7 people?



a. 1

b. 3

c. 11

d. 7

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Basic probability

How many ways can a committee of 5 be chosen from 16 people?

a. 4368

b. 8008

c. 4364

d. 4372

30 [Intermediate] • Target: Complement rule

How many ways can a committee of 3 be chosen from 14 people?

a. 364

b. 368

c. 1001

d. 360

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31 [Intermediate] • Target: Permutations and combinations

How many ways can a committee of 3 be chosen from 10 people?

a. 120

b. 116

c. 124

d. 210

32 [Intermediate] • Target: Independent events and the addition rule

How many ways can a committee of 4 be chosen from 9 people?

a. 126

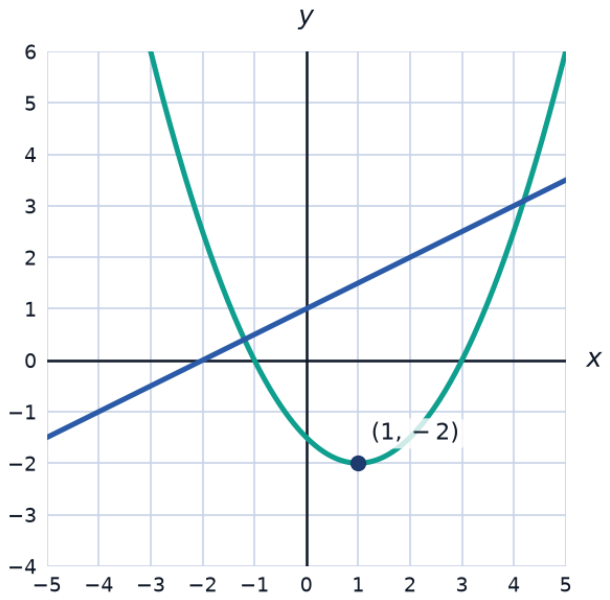
b. 122

c. 130

d. 132

33 [Intermediate] • Target: Discrete Random Variables & Variance

How many ways can a committee of 5 be chosen from 17 people?



a. 6192

b. 12376

c. 6188

d. 6184

34 [Intermediate] • Target: Continuous Distributions & Probabilities

How many ways can a committee of 2 be chosen from 14 people?

a. 87

b. 91

c. 364

d. 95

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35 [Intermediate] • Target: Conditional Probability & Joint Events

How many ways can 2 distinct positions be filled from 12 candidates, where order matters?

a. 66

b. 138

c. 132

d. 126

36 [Intermediate] • Target: Combinatorial Probability & Geometric Probability

How many ways can 4 distinct positions be filled from 14 candidates, where order matters?

a. 24018

b. 1001

c. 24030

d. 24024

37 [Intermediate] • Target: Basic probability

How many ways can 4 distinct positions be filled from 15 candidates, where order matters?

a. 32760

b. 32766

c. 32754

d. 1365

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Complement rule • Cross-Lesson: 3.01 Functions & Graphs

Events A and B are independent, with $P(A) = 0.6$ and $P(B) = 0.3$. Find $P(A \text{ and } B)$.

a. $\frac{23}{100}$

b. $\frac{13}{100}$

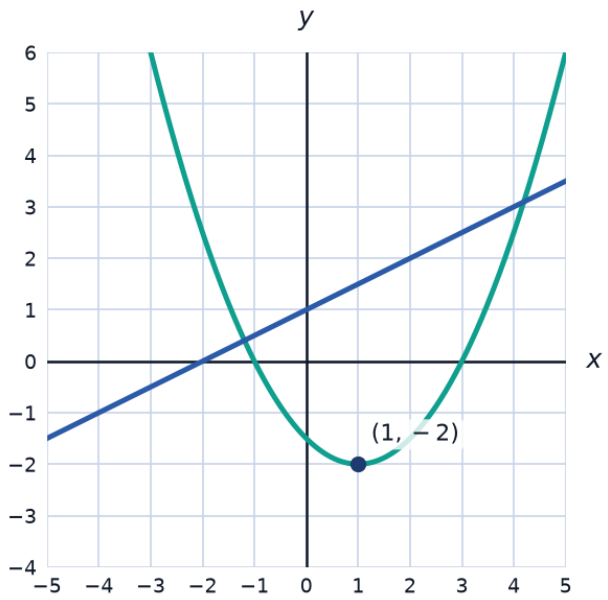
c. $\frac{9}{50}$

d. $\frac{9}{10}$

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39 [Challenging] • Target: Permutations and combinations • Cross-Lesson: 4.01 Matrices & Vectors

Out of 27 equally likely boxes, 7 are labeled, 8 are sealed, and 7 are both. One item is chosen at random. Find $P(\text{labeled or sealed})$.



a. $\frac{2}{9}$

b. $\frac{10}{27}$

c. $\frac{8}{27}$

d. $\frac{5}{9}$

40 [Challenging] • Target: Independent events and the addition rule • Cross-Lesson: 5.01 Introductory Calculus

Out of 37 equally likely numbers, 4 are even, 10 are prime, and 2 are both. One item is chosen at random. Find $P(\text{even or prime})$.

a. $\frac{14}{37}$

b. $\frac{10}{37}$

c. $\frac{12}{37}$

d. $\frac{15}{37}$

41 [Challenging] • Target: Discrete Random Variables & Variance • Cross-Lesson: 1.01 Complex Numbers

For the above variable X, what is the variance $\text{Var}(X)$?

a. 0.44

b. 0.50

c. 0.64

d. 0.84

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42 [Challenging] • Target: Continuous Distributions & Probabilities • Cross-Lesson: 2.01 Sequences & Series

A continuous random variable X is uniformly distributed over the interval $[2, 10]$. What is $P(4 \leq X \leq 7)$?

a. $\frac{3}{8}$

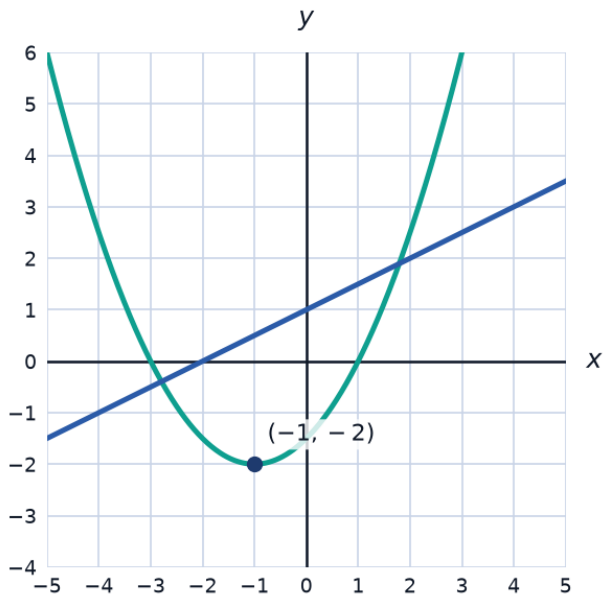
b. $\frac{1}{2}$

c. $\frac{5}{8}$

d. $\frac{1}{4}$

43 [Challenging] • Target: Continuous Distributions & Probabilities • Cross-Lesson: 3.01 Functions & Graphs

In any continuous probability distribution, what is $P(X = c)$ for any single exact point c?



a. 0

b. 1

c. $\frac{1}{2}$

d. Undefined

44 [Challenging] • Target: Conditional Probability & Joint Events • Cross-Lesson: 4.01 Matrices & Vectors

Given $P(A) = 0.6, P(B) = 0.3,$ and $P(A \cup B) = 0.7$, are events A and B independent?

a. No, because $P(A \cap B) = 0.20 \neq 0.18$

b. Yes, because $P(A \cap B) = 0.18$

c. Yes, because they are mutually exclusive

d. Cannot be determined

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45 [Challenging] • Target: Combinatorial Probability & Geometric Probability • Cross-Lesson: 5.01 Introductory Calculus

A point is chosen at random inside a square of side 10. What is the probability that it falls inside the inscribed circle?

a. $\frac{\pi}{4}$

b. $\frac{\pi}{2}$

c. $\frac{\pi}{8}$

d. $\frac{1}{2}$

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